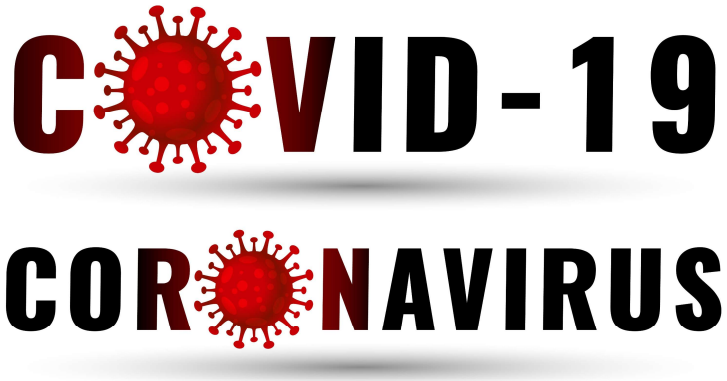
COVID – 19: Python Project

This project analyzes COVID-19 data using Python, Pandas, and Matplotlib. It studies confirmed, recovered, death, and active cases across different regions. Charts and graphs are used to visualize the data and understand the spread and impact of COVID-19.



About the Project:

A new column called **Active Cases** is created using the formula:
Active = Confirmed – Deaths – Recovered.

The project also groups the data by region to find the **total confirmed cases, deaths, and recoveries**. A **death rate** is calculated to see how serious the virus is in different regions.

Charts like **bar charts and scatter plots** are used to visualize the data. The analysis shows that **countries with more confirmed cases usually have more deaths**.

✅ **Overall, the project helps understand COVID-19 trends using data analysis and visualization.**

Questions:

1. Load the covid_19_data.csv dataset using pandas and display the first 5 rows.
2. Create a new column called Active Cases using the formula:
Active = Confirmed − Deaths − Recovered.
3. Find the country/region with the highest number of confirmed cases.
4. Check the dataset for missing values and display the total missing values in each column.
5. Sort the dataset by Confirmed cases in descending order and display the top 10 records.
6. Filter and display all records where Deaths are greater than 1000.
7. Display the column names and data types of the dataset.
8. Calculate the total number of confirmed cases worldwide.

9. Group the data by Region and calculate the total Confirmed, Deaths, and Recovered cases for each region.

10. Calculate the death rate for each region using the formula:

Death Rate = (Deaths / Confirmed) × 100

11. Display only the data for India.

12. Find the top 5 regions with the highest recovered cases.

13. Create a bar chart showing the top 10 regions with highest confirmed cases using Matplotlib.

14. Create a scatter plot comparing Confirmed cases and Deaths.

Thankyou